

ONEDRAFT

CAD-AI — Aria Powered

Complete Architectural & Implementation Stack Register — 1.0–50.0

Stack	Proposed document	Purpose
1.0	Foundational Technical Specification	Product definition, foundational principles, objectives, scope, architectural assumptions and core OneDraft requirements
2.0	System Architecture Specification	Overall OneDraft system architecture, subsystem boundaries, service relationships and architectural principles
3.0	MVP Technical Stack & Repository Architecture	Technology selection, repository structure, applications, packages, development environment and MVP implementation boundaries
4.0	API & Domain Schema Specification	Domain entities, service contracts, API resources, object relationships and foundational application schemas
5.0	Database Schema & OpenAPI Contract	PostgreSQL persistence model, database schemas, OpenAPI contract, versioning, transactions and API conventions
6.0	Executable Foundation Implementation Specification	Conversion of the architectural specification into executable foundation artifacts, domain models, migrations and service boundaries
7.0	Executable Application Stack & Runtime Architecture	Application execution model, runtime services, processes, service startup and application-layer architecture
8.0	Executable Software Stack & Foundation Implementation	Core software implementation structure, packages, dependencies, application foundation and executable components
9.0	Executable Application Stack & Implementation Architecture	Application implementation architecture, domain services, application services and execution boundaries
10.0	Executable Application Stack & Implementation Architecture	Consolidated executable application architecture and implementation progression toward the operational OneDraft platform
11.0	Executable Application & Domain Service Layer	Domain services, application services, command boundaries and executable business logic
12.0	Executable Repository Foundation & Core Application Runtime	Repository implementation, core runtime, application initialization, configuration and foundational execution
13.0	Application Runtime, Service Architecture & Execution Stack	Runtime service topology, service execution, internal interfaces and application process architecture
14.0	Computational Project Model & Geometry Engine	Computational representation of projects, architectural objects, relationships, geometry and parametric model behavior

Stack	Proposed document	Purpose
15.0	Executable Application Stack & Service Architecture	Executable service composition, application boundaries, internal service interfaces and operational implementation
16.0	Runtime & Application Orchestration Stack	Runtime orchestration, workflow coordination, service sequencing and execution control
17.0	Geometry & Parametric Model Engine	Parametric architectural geometry, spatial relationships, constraints, geometry computation and model regeneration
18.0	Compliance & Regulatory Engine	Jurisdictional code sources, regulatory rules, applicability analysis, code manifests and compliance computation
19.0	Validation & Verification Engine	Deterministic project validation, rule execution, verification results, severity classification and validation workflows
20.0	Validation & Coordination Engine	Cross-system coordination, model coordination, drawing coordination, constraint checking and multidisciplinary verification
21.0	Documentation & Drawing Generation Engine	Plans, elevations, sections, details, schedules, sheets, annotations, dimensions and architectural document generation
22.0	Project Review, Redline, Acceptance & Approval Engine	Customer review, redlines, revision requests, acceptance workflows, professional review and approval tracking
23.0	Runtime Orchestrator	Central orchestration of OneDraft workflows, subsystem execution, command routing and lifecycle coordination
24.0	Job & Worker Execution System	Asynchronous jobs, queues, workers, long-running operations, job state, retries and execution management
25.0	Artifact & Storage Management Engine	Generated files, document packages, geometry artifacts, object storage, hashes, references and artifact lifecycle
26.0	Integration & Event Bus	Internal events, asynchronous messaging, event contracts, service integration, event routing and system decoupling
27.0	Identity, Access & Security System	Identity foundations, authentication, organizational membership, access boundaries and security primitives
28.0	Identity, Security & Authorization System	Authorization policies, scopes, roles, permissions, resource access and security enforcement
29.0	Identity, Access Control & Security System	Consolidated access-control architecture, project authorization, tenant boundaries, privileged operations and security enforcement
30.0	Observability, Telemetry & System Health	Metrics, logs, traces, health checks, diagnostics, alerts and system-state visibility
31.0	Testing, Quality Assurance & Verification System	Unit, integration, contract, geometry, regression, E2E and acceptance testing
32.0	CI/CD, Build & Release Engineering	Automated builds, migrations, testing gates, deployments, versioning and releases
33.0	Infrastructure & Deployment Architecture	Containers, environments, networking, PostgreSQL, queues, workers, object storage and production topology
34.0	Configuration, Secrets & Environment Management	Environment separation, secrets, configuration hierarchy and runtime configuration

Stack	Proposed document	Purpose
35.0	Backup, Disaster Recovery & Business Continuity	Database recovery, artifact recovery, RPO/RTO, restore procedures and operational continuity
36.0	Data Lifecycle, Retention & Provenance System	Data ownership, retention, archival, lineage, hashes, immutable records and deletion rules
37.0	AI Intelligence & Model Provider Architecture	Aria orchestration, model adapters, prompts, context construction, model selection, evaluation and provider abstraction
38.0	AI Evaluation, Guardrails & Deterministic Command Verification	AI output evaluation, command safety, confidence, constraint preservation and deterministic validation
39.0	Standards, Architectural Graphics & CAD Interoperability	CAD standards, IFC, DXF/DWG interoperability, BIM exchange, layers, lineweights, symbols and graphic standards
40.0	Import, Export & External Data Exchange	Survey data, existing CAD/BIM, PDFs, images, specifications and external project data
41.0	Frontend & User Interaction Architecture	Project workspace, model viewer, drawing interface, redlines, review, acceptance and Aria interaction
42.0	Project Lifecycle & State Machine	Formal project states from intake through specification, modeling, validation, documentation, review and acceptance
43.0	Billing, Subscription & Entitlement Architecture	Plans, usage, quotas, feature entitlements, metering and organization billing boundaries
44.0	Multi-Tenant Organization & Project Isolation	Tenant boundaries, resource isolation, quotas, organization hierarchy and data segregation
45.0	API Gateway, Rate Limiting & Service Protection	Request controls, throttling, abuse protection, API quotas and service boundaries
46.0	Performance, Scalability & Capacity Architecture	Geometry workloads, concurrent projects, queue scaling, database scaling and performance targets
47.0	Disaster Testing, Security Testing & Operational Readiness	Failure injection, recovery tests, penetration/security testing and production-readiness gates
48.0	Developer Platform, SDK & Extension Architecture	Python/TypeScript clients, SDKs, plugins, external integrations and future third-party extensions
49.0	Product Configuration & Administrative Control Plane	Platform administration, feature flags, jurisdiction configuration, templates and controlled system settings
50.0	OneDraft End-to-End System Integration Specification	Master integration document proving how all preceding subsystems operate together as one coherent OneDraft platform

STACK PROGRESSION

1.0–5.0

FOUNDATION



6.0–16.0

EXECUTABLE APPLICATION FOUNDATION



17.0-22.0
CORE COMPUTATIONAL & DOCUMENTATION ENGINES
↓
23.0-29.0
RUNTIME, JOBS, ARTIFACTS, EVENTS & SECURITY
↓
30.0-36.0
OPERATIONS, QUALITY, INFRASTRUCTURE & DATA GOVERNANCE
↓
37.0-40.0
AI, CAD STANDARDS & DATA EXCHANGE
↓
41.0-49.0
USER PLATFORM, LIFECYCLE, COMMERCIAL & SCALE SYSTEMS
↓
50.0
END-TO-END SYSTEM INTEGRATION

ARCHITECTURAL COMPLETION STATE

FOUNDATION
+
DOMAIN
+
COMPUTATIONAL PROJECT MODEL
+
GEOMETRY
+
COMPLIANCE
+
VALIDATION
+
DOCUMENTATION
+
REVIEW / ACCEPTANCE
+
RUNTIME ORCHESTRATION
+
JOB EXECUTION
+
ARTIFACT STORAGE
+
EVENT BUS
+
IDENTITY / SECURITY
+
OBSERVABILITY
+
TESTING
+
CI/CD
+
INFRASTRUCTURE
+
DATA GOVERNANCE
+
ARIA INTELLIGENCE
+

CAD INTEROPERABILITY
+
EXTERNAL DATA EXCHANGE
+
FRONTEND
+
PROJECT LIFECYCLE
+
BILLING
+
MULTI-TENANCY
+
API PROTECTION
+
SCALABILITY
+
OPERATIONAL READINESS
+
SDK / EXTENSIONS
+
ADMINISTRATION
↓
ONEDRAFT END-TO-END SYSTEM

MASTER STACK REGISTER

1.0–50.0 — ONEDRAFT CAD-AI / ARIA POWERED

The 50-stack architecture establishes the complete progression from foundational technical definition through executable software, computational modeling, regulatory analysis, documentation generation, runtime execution, security, operations, artificial intelligence, user interaction, commercialization, scalability and final end-to-end system integration.

The architecture therefore proceeds from **specification** → **implementation** → **computation** → **orchestration** → **operation** → **integration**, with each stack inheriting the contracts and system boundaries established by the preceding stacks.